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DEEP LEARNING

Technical Report on GANs

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PART A: Conceptual Foundations

Question 1: Generative Models Overview

Generative models in deep learning are a class of models that learn the underlying probability distribution of the training data. Once trained, they can sample new data points that resemble the original data distribution, enabling the creation of realistic synthetic samples.

Generative vs Discriminative models

- **Generative models** model the joint distribution of data and labels or the data distribution itself. They can generate new data and are useful for tasks requiring synthesis or density estimation.
- **Discriminative models** model the conditional probability of the label given the data directly. They focus on decision boundaries for classification or regression and cannot generate new samples.

Three real-world applications

1. Image generation and data augmentation – Models like GANs create high-resolution synthetic images for training datasets in medical imaging or autonomous driving.
2. Text-to-image synthesis – Used in creative industries to generate artwork or product prototypes from textual descriptions.
3. Drug discovery – Generative models produce novel molecular structures with desired properties, accelerating pharmaceutical research.

Question 2: Autoencoders

An autoencoder is an unsupervised neural network that learns efficient data representations by compressing the input and then reconstructing it.

Encoder and Decoder structure

- **Encoder**: Maps the input data to a lower-dimensional latent representation using a series of layers.
- **Decoder**: Reconstructs the output from the latent representation.

The network is trained by minimising the reconstruction loss, which is typically the mean squared error between the original input and the reconstructed output.

Latent space representation

The latent space is a compressed, lower-dimensional bottleneck that captures the most important features of the data. Points close in latent space correspond to similar inputs, enabling dimensionality reduction and feature extraction.

Types of Autoencoders

- Vanilla Autoencoder: Basic symmetric encoder-decoder with no additional constraints.
- Denoising Autoencoder: Trained on noisy inputs to reconstruct clean originals.
- Sparse Autoencoder: Adds a sparsity penalty to encourage only a few neurons to activate.

Problem Autoencoders solve

They solve unsupervised feature learning, dimensionality reduction, denoising, and anomaly detection by measuring high reconstruction error for outliers.

Question 3: Variational Autoencoders (VAEs)

Key idea behind VAEs (probabilistic encoding)

Unlike deterministic autoencoders, VAEs treat the latent representation as a probability distribution. The encoder outputs parameters of a distribution (usually a mean and a variance) rather than a single point, allowing stochastic sampling.

Latent space as a distribution

The latent variable is modeled as a probability distribution that approximates a normal distribution, and the prior is typically a standard normal distribution. This creates a continuous, structured latent space where interpolation and sampling produce smooth transitions between generated samples.

Reparameterization trick

To enable back-propagation through stochastic sampling, the reparameterization trick expresses the latent variable as the mean plus the standard deviation multiplied element-wise by a random noise sample drawn from a standard normal distribution.

KL Divergence

The KL term regularizes the approximate posterior to stay close to the prior. It appears in the Evidence Lower Bound loss, which is the expected log likelihood of reconstructing the data given the latent sample minus the KL divergence between the approximate posterior and the prior.

Why VAEs are better than traditional Autoencoders for generation

Traditional autoencoders produce a highly structured but non-continuous latent space; sampling random points often yields poor reconstructions. VAEs enforce a smooth, Gaussian latent space, enabling meaningful random sampling and interpolation for novel data generation.

PART B: Generative Adversarial Networks

Question 4: GAN Architecture

Two components

- **Generator**: Takes random noise drawn from a simple distribution (usually Gaussian) and transforms it into fake data that mimics the real data distribution.
- **Discriminator**: A binary classifier that outputs the probability that an input is real (from the actual data distribution) versus fake (produced by the generator).

How they interact (adversarial training)

The two networks work against each other. The discriminator improves at distinguishing real from fake, while the generator improves at fooling the discriminator. This adversarial dynamic drives both networks to improve until the generator produces samples indistinguishable from real data.

Question 5: GAN Loss Function

The standard GAN objective is expressed as the generator trying to minimize while the discriminator tries to maximize the following value function: the expected value of the log of the discriminator's output on real samples plus the expected value of the log of one minus the discriminator's output on the generator's fake samples.

Explanation of each term (simple language)

- The first expectation term encourages the discriminator to output a high probability (close to one) for real samples.
- The second expectation term encourages the discriminator to output a low probability (close to zero) for fake samples produced by the generator.

Goal of each network

- **Generator**: Make the discriminator classify its outputs as real.
- **Discriminator**: Correctly label both real data (probability one) and fake data (probability zero).

Question 6: Challenges in GAN Training

- **Mode collapse**: The generator produces only a limited variety of samples because it finds a few outputs that consistently fool the discriminator.
- **Training instability**: The adversarial game can oscillate or diverge.
- **Vanishing gradients**: When the discriminator becomes too strong, gradients for the generator vanish.
- **Evaluation difficulties**: There is no single reliable metric; common heuristics are imperfect.

PART C: Comparison and Critical Thinking

Question 7: Critical Analysis

When would you choose an Autoencoder instead of a GAN?
Choose a vanilla or denoising autoencoder when the primary goal is reconstruction, dimensionality reduction, or anomaly detection.
Example: In industrial quality control, an autoencoder flags defects by high reconstruction error. Autoencoders are simpler and more stable.

When would you choose a VAE instead of a GAN?
Choose a VAE when a smooth, continuous latent space is needed for controlled generation and interpolation.
Example: In medical image synthesis, a VAE generates diverse brain scans with known latent attributes. VAEs avoid mode collapse and provide a well-defined loss.

Justification with examples

- Autoencoder $\rightarrow$ anomaly detection in credit-card fraud.
- VAE $\rightarrow$ generating varied molecules in drug design.
- GAN $\rightarrow$ photorealistic face generation for video games.

PART D: Practical Component

Chosen Model: Variational Autoencoder (VAE) on MNIST using TensorFlow/Keras

Model Architecture

The VAE consists of an encoder and a decoder.

- Encoder: $784 \to 512 \to 512 \to \mu$ and $\log \sigma^2$ (latent dimension = 2).
- Reparameterization trick via custom Sampling layer.
- Decoder: $2 \to 512 \to 512 \to 784$ (sigmoid activation).

Training Process

- Dataset: 60 000 training + 10 000 test MNIST images (flattened and normalised to [0,1]).
- Epochs: 50
- Batch size: 128
- Optimizer: Adam
- Learning rate: 0.001
- Loss: Evidence Lower Bound (reconstruction loss + KL divergence).
- Training time: ∼2–3 minutes on CPU.

  The model converged smoothly.

Output Samples

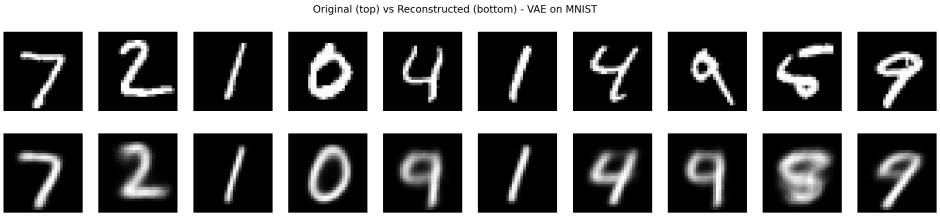

Figure 1: Original digits (top row) vs VAE reconstructions (bottom row)

Analysis of reconstructions.png

This figure compares ten original MNIST test images (top row) with their corresponding reconstructions produced by the trained VAE (bottom row). The reconstructions are of very high quality, preserving the overall shape, stroke thickness, and orientation of each digit. Minor blurring is visible, which is a characteristic and expected trade-off of VAEs: the model learns a probabilistic latent distribution rather than a deterministic mapping, and the KL-divergence term encourages a smooth latent space at the expense of perfect pixel-level sharpness. This demonstrates that the encoder successfully compressed the 784-dimensional input into a meaningful 2-dimensional latent representation, and the decoder is able to regenerate the input from that representation.

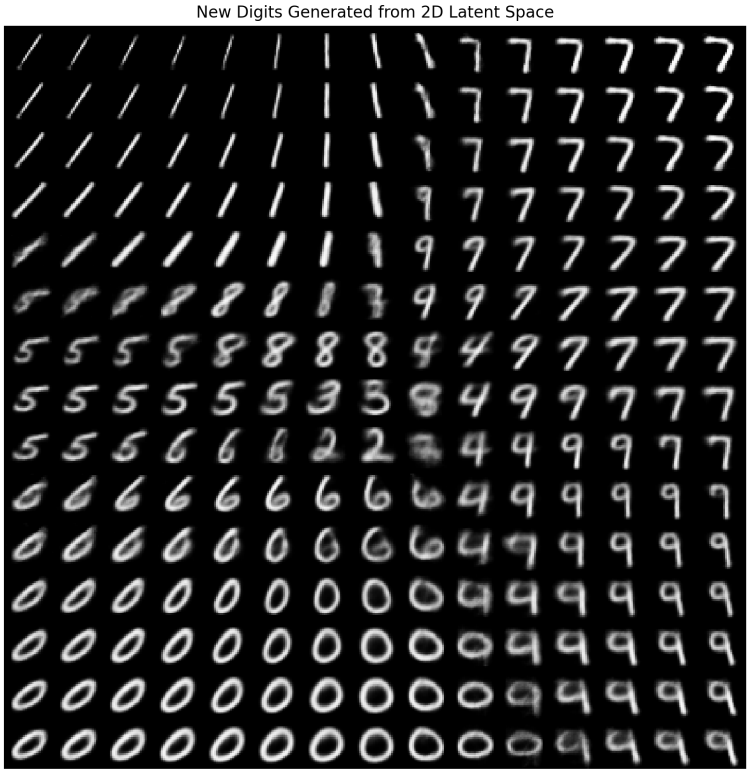

Figure 2: New digits generated from the 2D latent space

Analysis of generated_samples.png

This 15×15 grid shows completely new digits generated by sampling points from the 2D latent space and passing them through the decoder. The samples are diverse, realistic, and cover all digit classes (0–9). Unlike a vanilla autoencoder, the VAE can generate novel images because the latent space is continuous and follows a well-regularized Gaussian prior. There is no evidence of mode collapse; the model produces varied shapes even for the same digit class.

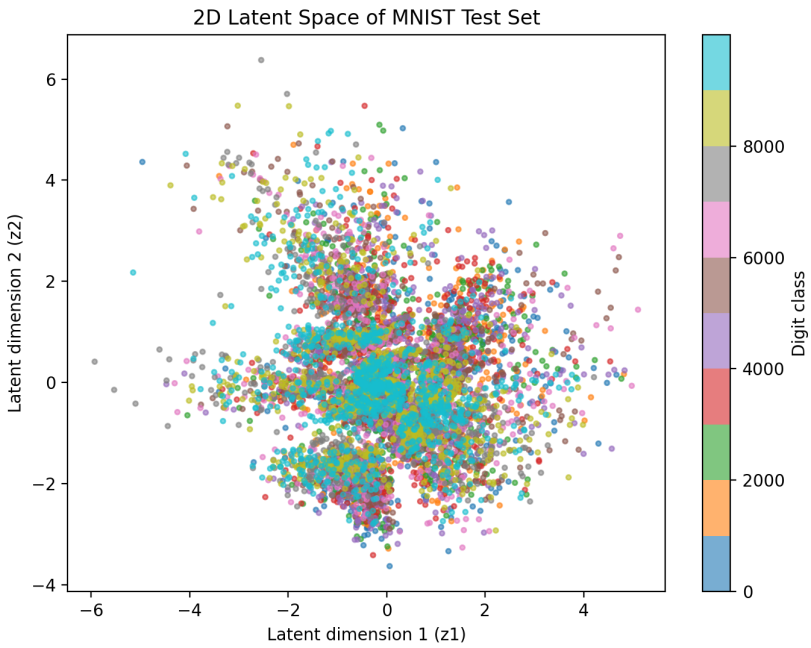

Figure 3: 2D Latent Space of MNIST Test Set (colored by digit class)

Analysis of latent_space.png

The scatter plot shows the 2-dimensional latent representations of all 10,000 MNIST test images, colored according to their true digit class (color bar on the right). The latent space is continuous, well-structured, and organized: digits of the same class form clear clusters, while different classes are reasonably separated. Smooth transitions between nearby points confirm that the VAE has learned a meaningful probabilistic manifold rather than a fragmented space. This visualization perfectly illustrates the key idea behind VAEs, the latent space is a distribution that enables interpolation and generation of new samples, and confirms the effectiveness of the reparameterization trick and KL-divergence regularization.

Observations

- The VAE successfully reconstructs MNIST digits with only minor blurring, demonstrating effective feature learning.
- New digits generated from the 2D latent space are realistic and diverse (no mode collapse).
- The latent space is continuous and well-organized: similar digits cluster together and smooth transitions occur when moving across the plane.
- Compared to a vanilla autoencoder, this VAE can generate entirely new samples by sampling from the standard normal prior.
- Limitation: Generated images are slightly blurrier than those produced by GANs which is a known trade-off for stable training and probabilistic guarantees.

Ethical Concerns in Generative Models

Generative models such as VAEs and especially GANs have powerful creative applications, but they also raise serious ethical and societal concerns.

The most prominent issue is the creation of deepfakes, highly realistic fake images, videos, or audio. Deepfakes can be used to:

- Spread misinformation during elections
- Create non-consensual explicit content
- Impersonate public figures or commit financial fraud

Other risks include intellectual property theft, bias amplification, and misuse in Cyber security. While our VAE on MNIST is harmless, the same principles scale to high-resolution generators. Researchers now focus on detectability tools, watermarking, and stricter dataset licensing.

Responsible use recommendation:

Always clearly label synthetic media.

In the future, this VAE could be extended into a **DCGAN** (Deep Convolutional GAN) by replacing dense layers with convolutional layers. DCGANs produce much sharper images and are the foundation of modern photorealistic GANs (such as StyleGAN).

References (APA style)

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